

TOKAT, Turkey

WMO: 170860

Lat: 40.3311N

Lon: 36.5578E

Elev: 608

StdP: 94.23

Time Zone: 3.00 (E03)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-6.3	-13.1	1.3	-6.6	-10.9	1.6	-3.5	8.4	11.1	7.0	7.8	1.8	90	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	34.7	19.8	32.6	19.3	30.8	18.9	21.5	31.7	20.6	30.0	19.8	28.5	3.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	14.1	25.8	17.3	13.3	25.1	16.6	12.7	24.5	65.6	31.5	62.3	30.3	59.4	28.6	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.9	5.2	DB	-12.2	38.7	4.1	1.4	-15.1	39.7	-17.5	40.5	-19.8	41.2	-22.8	42.2	(4)
(5)			WB	-12.7	22.6	3.9	1.5	-15.6	23.7	-17.9	24.6	-20.1	25.4	-22.9	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	2.8	4.7	8.5	12.9	17.2	20.8	23.2	24.0	20.1	15.0	8.3	3.6	(6)
(7)		DBStd	8.39	4.85	5.09	4.56	4.12	3.57	2.72	2.81	2.72	3.36	3.70	3.93	4.66	(7)
(8)		HDD10.0	773	226	158	84	18	1	0	0	0	0	6	78	202	(8)
(9)		HDD18.3	2301	483	383	305	166	64	8	1	1	21	112	302	457	(9)
(10)		CDD10.0	2041	2	8	38	106	225	324	408	434	304	162	26	4	(10)
(11)		CDD18.3	527	0	0	1	4	29	82	150	176	74	10	0	0	(11)
(12)		CDH23.3	5126	0	0	11	86	349	777	1449	1655	694	104	1	0	(12)
(13)		CDH26.7	1971	0	0	2	15	101	250	619	728	241	17	0	0	(13)
(14)	Wind	WSAvg	2.2	2.0	2.3	2.4	2.4	2.2	2.2	2.5	2.2	1.9	1.6	1.8	(14)	
(15)	Precipitation	PrecAvg	407	45	33	36	54	56	38	11	5	18	35	42	51	(15)
(16)		PrecMax	517	116	83	86	104	109	93	68	36	50	141	114	103	(16)
(17)		PrecMin	313	6	4	2	13	17	3	0	0	1	0	1	9	(17)
(18)		PrecStd	60	29	19	21	25	23	21	15	8	11	31	28	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	19.1	24.7	28.6	32.5	33.9	37.5	37.2	34.8	28.7	21.8	16.7	(19)
(20)			MCWB	8.0	9.7	12.3	14.1	18.0	19.0	20.2	20.9	17.7	16.2	12.0	9.0	(20)
(21)		2%	DB	12.9	16.8	21.1	25.9	29.5	31.4	34.6	34.8	31.6	26.0	19.2	14.2	(21)
(22)			MCWB	6.8	8.8	10.8	13.2	16.8	18.5	20.1	20.3	17.4	15.5	11.5	8.1	(22)
(23)		5%	DB	11.3	14.3	18.6	23.4	27.3	29.5	32.1	32.6	29.3	24.0	17.0	12.1	(23)
(24)			MCWB	5.9	7.7	9.6	12.2	15.9	18.2	19.6	20.0	17.0	14.8	10.2	7.0	(24)
(25)		10%	DB	9.6	12.4	16.5	21.1	25.0	27.7	30.1	30.7	27.4	21.9	14.9	10.3	(25)
(26)			MCWB	5.2	6.6	8.9	11.6	15.3	17.6	18.8	19.5	16.6	14.1	9.3	6.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	10.8	13.1	16.2	19.6	21.1	22.9	22.7	20.8	17.9	13.4	9.8	(27)
(28)			MCDWB	14.1	17.0	22.7	25.8	30.2	30.4	33.5	34.1	30.9	25.8	19.6	14.4	(28)
(29)		2%	WB	7.2	9.1	11.2	14.2	17.9	19.9	21.5	21.6	19.1	16.3	12.0	8.7	(29)
(30)			MCDWB	11.6	15.1	19.4	22.8	27.4	28.6	31.6	32.3	27.4	23.3	17.8	12.9	(30)
(31)		5%	WB	6.4	8.0	10.1	13.1	16.7	19.1	20.5	20.8	18.1	15.5	10.7	7.6	(31)
(32)			MCDWB	10.3	13.7	17.2	21.6	25.2	27.1	29.7	30.4	26.4	22.2	15.4	11.5	(32)
(33)		10%	WB	5.6	6.9	9.1	12.0	15.6	18.2	19.6	20.0	17.2	14.6	9.7	6.6	(33)
(34)			MCDWB	9.2	11.9	15.5	19.5	23.3	25.9	27.9	28.9	25.1	20.7	14.3	9.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	8.5	10.0	12.1	12.8	12.7	12.8	12.9	13.2	11.1	10.0	7.0	(35)
(36)			MCDBR	9.2	11.6	13.8	16.4	16.6	15.5	16.4	16.0	16.4	14.7	13.0	9.2	(36)
(37)			MCWBR	5.0	5.8	6.6	7.1	6.8	5.7	5.6	5.4	5.8	6.1	6.6	5.1	(37)
(38)		5% WB	MCDBR	8.0	10.7	12.4	14.3	15.1	13.7	14.4	14.3	13.9	12.4	11.5	8.5	(38)
(39)			MCWBR	4.7	5.5	6.3	6.8	6.9	5.5	5.9	5.6	5.6	5.6	6.1	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.354	0.402	0.442	0.414	0.384	0.385	0.393	0.365	0.374	0.334	0.318	(40)	
(41)		taud	2.315	2.235	2.151	2.083	2.241	2.358	2.368	2.331	2.405	2.339	2.411	2.382	(41)	
(42)		Ebn at Noon	829	847	842	833	866	890	886	870	875	818	811	803	(42)	
(43)		Edh at Noon	90	113	137	157	137	122	120	121	105	101	82	78	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.96	2.89	3.68	4.84	5.87	6.73	6.80	6.17	4.92	3.29	2.27	1.65	(44)	
(45)		RadStd	0.23	0.39	0.36	0.58	0.54	0.56	0.40	0.37	0.34	0.29	0.29	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (6 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+85	+59		

Nomenclature: See separate page